

Python Code

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1  #Blackjack
2  import random
3
4  #global variables
5  Computer_Total = 0
6  Player_Total = 0
7  Player = "A"
8
9  # deck of cards
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15  cards =
["1", "1", "1", "1", "2", "2", "2", "2", "3", "3", "3", "3", "4", "4", "4", "4", "5", "5",
16   "5", "5", "6", "6", "6", "6", "7", "7", "7", "7", "8", "8", "8", "8", "9", "9", "9", "9", "1
0", "10", "10", "10"]
17
18
19  # Displays the intro instructions for the player
20  def Intro():
21      print("Welcome to Blackjack!\nDraw cards and try to reach a total of 21 –
or get as close as possible without going over.\nOn your turn, type 'Hit' to draw a
card, or 'Pass' to end your turn.\n(Note: Only numbered cards, no face cards.)")
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23
24  # Randomly draws a card from the deck and removes it
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31  def Hit(cards):
32      random_card = random.choice(cards)
33      cards.remove(random_card)
34      return int(random_card)
35
36
37  # Switches the current player (between human and computer)
38  def Pass():
39      global Player
40      if Player == "A":
41          Player = "Computer"
42      else:
43          Player = "A"
44
45
46  # Function to ask the current player if they want to hit or pass
47  def option():
48      global Player_Total
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49     first = Hit(cards)
50     second = Hit(cards)
51     Player_Total += first + second
52     print("\nYour first two cards are: ", first, second)
53     print("Player Total: ", Player_Total)
54     choice = input("\nPlayer " + Player + "\nWould you like to Hit or Pass?
")
55     return choice
56
57
58 # Handles the player's turn, input, and action
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66 def action(player_input):
67     global Player_Total
68     global cards
69     while Player_Total < 21:
70         try:
71             if player_input.lower() == "hit" or player_input.lower() == "h":
72                 card = Hit(cards)
73                 Player_Total += card
74                 print("Card value:" + str(card))
75                 print("Total = " + str(Player_Total))
76                 if Player_Total == 21:
77                     print("Good job, you hit 21!")
78                     Pass()
79                     break
80                 elif Player_Total > 21:
81                     print("You went over 21!")
82                     Pass()
83                     break
84             elif player_input.lower() == "pass" or player_input.lower() ==
"p":
85                 print("You passed your turn")
86                 print("Player total: ", Player_Total)
87                 Pass()
88                 break
89             else:
90                 raise ValueError("Invalid input")
91             player_input = input("\nWould you like to Hit or Pass? ")
92         except Exception as e:
93             print("You can only choose from Hit or Pass", e)
94             player_input = input("\nWould you like to Hit or Pass? ")
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102 # Function for the computer's turn – it keeps hitting until the total is at 1
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103 def computer():
104     global Computer_Total
105     global Player_Total
106     print("\n-----Computer's Turn-----")
107     while Computer_Total < 17:
108         draw = Hit(cards)
109         Computer_Total += draw
110         print("\nComputer drew: " + str(draw))
111         print("Computer Total: " + str(Computer_Total))
112     print("Player total: ", Player_Total)
113
114
115 # Function to check who wins based on player and computer totals
116 def check_win():
117     global Player_Total
118     global Computer_Total
119     if Player_Total <= 21:
120         if Computer_Total <= 21:
121             if Player_Total > Computer_Total:
122                 print("Congratulation, You win!!")
123             elif Player_Total < Computer_Total:
124                 print("You lost!!")
125             elif Player_Total == Computer_Total:
126                 print("You lost!! Computer wins draws :(")
127         else:
128             print("You Win!!")
129     else:
130         print("You Lost!!")
131
132
133 # Function to run the full game flow: intro, player turn, computer turn, win
check
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152 def play_game():
153     Intro()
154     player_choice = option()

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155     action(player_choice)
156     computer()
157     check_win()
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172 play_game()
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174
175 #hgfdc
```